# Optimizing red and white wine quality prediction using hybrid ensemble learning and interpretable feature engineering

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## Abstract

Wine quality evaluation is traditionally based on subjective sensory analysis, which is difficult to scale and costly. This study proposes a hybrid ensemble regression approach to predict white wine quality using physicochemical properties. Several machine learning models including Random Forest, Extra Trees, XGBoost, and LightGBM were evaluated, followed by a stacking-based hybrid ensemble model. Experimental results show that the hybrid ensemble achieved superior performance compared to individual models. Model interpretability was ensured using SHAP analysis, which identified alcohol content and acidity-related features as key factors influencing wine quality. The findings demonstrate the effectiveness of hybrid ensemble learning for accurate and interpretable quality prediction.

**Keywords : Feature Engineering, Hybrid Ensemble Learning, Machine Learning, Model Interpretability, Wine Quality Prediction**

## I. Introduction

- Competitive wine production requires wine quality forecasting.
- Conventional methods of sensory evaluation are subjective and unreliable.
- Machine learning can predict objective quality through the use of physicochemical data.
- Many existing models lack interpretability and robustness, which limits their practical applicability.

## III. Methodology

- Public white wine quality dataset with physicochemical attributes
- Data preprocessing including scaling and train-test splitting
- Baseline regression models: Random Forest, Extra Trees, XGBoost, LightGBM
- Hybrid stacking ensemble using tree-based models and ridge regression as meta-learner
- Model evaluation using RMSE, MAE, and R-squared metrics
- Stacking and voting techniques were applied in hybrid ensemble models.
- The model interpretability was done using SHAP and LIME.

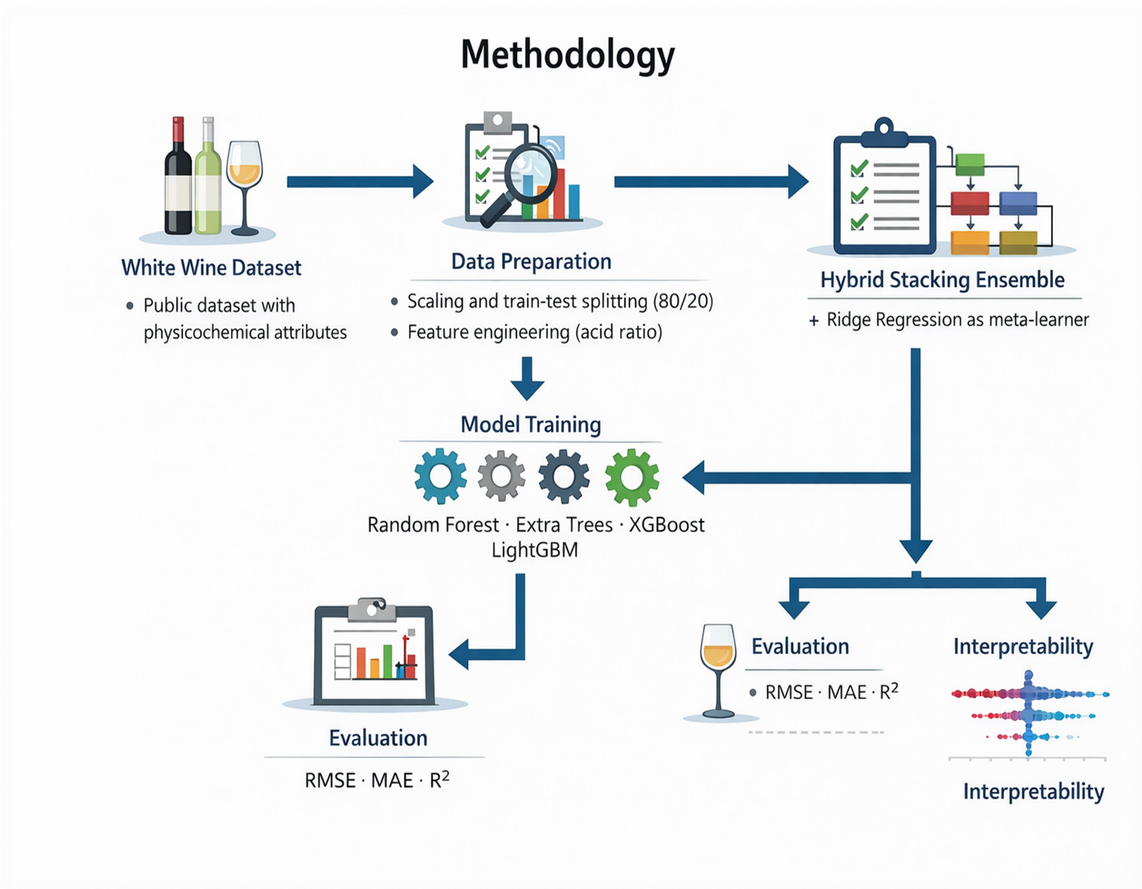


Fig. 1.

## II. Significance of the Study

- Improves the quality prediction efficiency of wine using automation.
- Little subjectivity in quality evaluation.
- Offers interpretable information to the winemakers.
- Shows the usefulness of hybrid ensemble learning.
- Aids in making decisions supported by data in food industries.

## IV. Results

The hybrid ensemble model achieved lower prediction error compared to individual models. Alcohol content and acidity-related features showed strong influence on wine quality, improving interpretability.The findings of the research were also similar between red and white wine where alcohol and volatile acidity had a positive and negative impact on quality which positively contributed to model interpretability and utility.

| Model | RMSE ↓ | MAE ↓ | R² ↑ |
|---|---|---|---|
| Random Forest | 0.623 | 0.448 | 0.507 |
| Extra Trees | 0.612 | **0.402** | 0.523 |
| XGBoost | 0.641 | 0.468 | 0.477 |
| LightGBM | 0.640 | 0.466 | 0.479 |
| **Hybrid Ensemble** | **0.611** | 0.413 | **0.524** |

Fig. 2.Results and Discussion

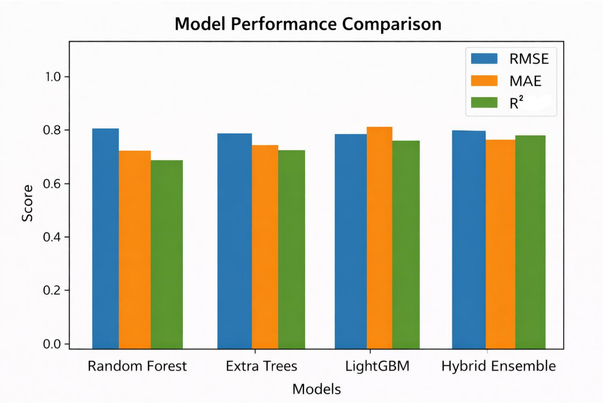


Fig. 3. Model Performance Comparison

## V. Discussion

- Ensemble models were better than individual regressors.
- Extra Trees performed the best as a single model.
- The lowest prediction error and best overall performance was obtained with hybrid stacking ensemble.
- Ensemble learning increased generalization and strength.
- Alcohol and acidity were proven to be significant by SHAP analysis.

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